

SIMATS ENGINEERING

ASSESSMENT TOOL 1: Class Test

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1506

COURSE OUTCOME COVERED

CO4: *Analyze big data characteristics and challenges across domains including security and application aspects.*
(BL4)

Assessment Tool Weightage: 40%

Dave's Taxonomy: Articulation (Level 4)
Question Count: 10

Total Marks: 30

Code of Conduct

I M. Pranay (192525382)_ certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: M. Pranay_____

Assessment Tasks

Class Test

1. Analyze a multinational retail organization struggling with fragmented customer data across online and offline channels, and propose a unified big data architecture that enables real-time customer analytics and personalized recommendations.
2. Evaluate the effectiveness of distributed computing models in processing large-scale scientific research data, and justify how parallel processing improves computational efficiency and scalability compared to centralized systems.
3. Design a fraud detection framework for a digital payment platform using big data analytics, and justify how streaming data processing and predictive models improve fraud prevention accuracy.
4. Assess the challenges of storing and processing multimedia data such as video, audio, and images in a big data environment, and recommend suitable technologies and storage mechanisms for efficient management.
5. Construct a disaster recovery and business continuity strategy for a cloud-based big data platform, and justify how redundancy, replication, and automated failover mechanisms ensure operational resilience.
6. Analyze the impact of data quality issues such as inconsistency, duplication, and missing values in a large-scale analytics platform, and propose data governance strategies to improve analytical reliability.
7. Design a scalable recommendation system for an online streaming service using big data technologies, and justify how user behavior analytics and distributed processing enhance recommendation accuracy and user engagement.
8. Evaluate the role of cloud computing in modern big data ecosystems, and critically examine its advantages and limitations in terms of scalability, cost management, flexibility, and security.

9. Develop a real-time analytics framework for transportation and logistics management, and justify how big data technologies improve route optimization, fuel efficiency, and operational decision-making.
10. Critically analyze ethical and legal challenges associated with big data collection and analytics, and recommend organizational policies that ensure responsible data usage, regulatory compliance, and user trust.

Class Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Concept Accuracy	30%	Accurate and well-expressed technical answers	Mostly accurate answers	Partial accuracy	Inaccurate answers
Coverage	20%	Covers all required points	Covers most points	Limited coverage	Very poor coverage
Use of Examples	15%	Uses relevant and meaningful examples	Some relevant examples	Weak examples	No useful examples
Analytical Awareness	20%	Shows strong awareness of issues and implications	Reasonable awareness	Basic awareness	Minimal awareness
Clarity	15%	Clear and concise writing	Generally clear	Some unclear expressions	Poor clarity

ANSWERS

1. Unified Big Data Architecture for Retail

A multinational retail organization can integrate online and offline customer data using a **unified big data architecture**.

- **Data Sources:** Websites, mobile apps, POS systems, CRM, social media and IoT devices.
- **Data Ingestion:** Apache Kafka or cloud-based ingestion services collect real-time data.
- **Data Storage:** A data lake such as Amazon S3, Azure Data Lake or Hadoop HDFS stores structured and unstructured data.
- **Data Processing:** Apache Spark processes large datasets in batch and real time.
- **Analytics Layer:** Customer behavior, purchase history and browsing patterns are analyzed.
- **Recommendation Engine:** Machine learning models generate personalized product recommendations.
- **Visualization:** Dashboards display customer trends and sales information.
- **Security:** Encryption, authentication and access control protect customer data.

Example: When a customer browses a product online, the system can immediately analyze their previous purchases and recommend related products.

2. Distributed Computing for Scientific Research

Distributed computing divides large scientific workloads among multiple computers.

- Large datasets are divided into smaller partitions.
- Multiple processing nodes execute tasks simultaneously.
- Frameworks such as **Hadoop and Apache Spark** support distributed processing.
- **Parallel processing** reduces execution time.
- Distributed systems can scale by adding more computing nodes.
- They provide better fault tolerance than a single centralized system.
- Scientific applications such as weather prediction, genome analysis and astronomy benefit from distributed computing.

Comparison:

Centralized Computing	Distributed Computing
Single/main processing system	Multiple processing nodes
Limited scalability	Highly scalable
Longer processing time	Faster parallel processing
Single point of failure	Fault tolerance
Suitable for smaller workloads	Suitable for huge datasets

Thus, distributed computing provides better **efficiency, scalability and reliability** for scientific research.

3. Big Data Fraud Detection Framework

A digital payment platform can use real-time big data analytics to detect fraudulent transactions.

Framework:

Transaction → Data Ingestion → Stream Processing → Feature Extraction → ML Model → Fraud Score → Alert/Block

- Payment transactions are continuously collected.
- **Apache Kafka** can ingest streaming transaction data.
- **Spark Streaming/Flink** processes transactions in real time.
- Features such as transaction amount, location, device and transaction frequency are extracted.
- Machine learning models identify suspicious behavior.
- A fraud score is generated for each transaction.
- High-risk transactions can be blocked or sent for verification.
- Historical data can be used to continuously improve the predictive model.

Example: If a card is used in Chennai and immediately afterward in another distant country, the system can identify the unusual behavior and generate an alert.

Streaming analytics provides **faster detection**, while predictive models improve **fraud detection accuracy**.

4. Multimedia Big Data Storage and Processing

Multimedia data includes videos, images and audio files that are large, complex and often unstructured.

Challenges:

- Very large file sizes.
- High storage requirements.
- Different file formats.
- High bandwidth requirements.
- Difficult indexing and searching.
- High computational requirements for processing.
- Maintaining data quality and metadata.

Suitable technologies:

- **Hadoop HDFS** for distributed storage.
- **Cloud object storage** such as Amazon S3 or Azure Blob Storage.
- **Apache Spark** for large-scale processing.
- NoSQL databases for metadata.
- Compression techniques to reduce storage requirements.
- CDN technology for faster multimedia delivery.

A combination of **distributed storage, compression, metadata management and parallel processing** enables efficient multimedia management.

5. Disaster Recovery and Business Continuity

A cloud-based big data platform requires a disaster recovery strategy to maintain availability during failures.

Strategy:

- Store multiple copies of important data.
- Use **data replication** across different availability zones or regions.
- Maintain redundant servers and storage systems.
- Perform regular automated backups.
- Use automated failover mechanisms.
- Monitor infrastructure continuously.
- Define **RTO** and **RPO** requirements.
- Periodically test disaster recovery procedures.

Example: If the primary cloud region becomes unavailable, the system can automatically switch to a replicated secondary region.

Redundancy prevents dependence on a single component, **replication** protects data, and **automated failover** reduces downtime.

Therefore, these mechanisms provide **high availability, reliability and business continuity**.

6. Impact of Data Quality Issues

Poor data quality can significantly reduce the reliability of big data analytics.

Major problems:

- **Inconsistency:** Different systems may store different values for the same customer.
- **Duplication:** Repeated records can produce incorrect statistics.
- **Missing values:** Incomplete information can affect machine learning models.
- **Incorrect data:** Wrong values lead to inaccurate decisions.
- **Outdated data:** Old information may not represent current conditions.

Data governance strategies:

- Establish data quality standards.
- Perform data cleansing and validation.
- Remove duplicate records.
- Handle missing values using appropriate techniques.
- Maintain master data management.
- Define data ownership and responsibilities.
- Monitor data quality continuously.
- Use metadata and data lineage.
- Apply access control and security policies.

Good data governance improves **accuracy, consistency, trust and analytical reliability**.

7. Scalable Recommendation System

An online streaming service can build a recommendation system using big data technologies.

Architecture:

User Activity → Data Collection → Distributed Storage → Data Processing → ML Model → Recommendations

- Collect viewing history, ratings, searches and watch duration.
- Store large-scale user data in a data lake.
- Use **Kafka** for real-time activity collection.
- Use **Apache Spark** for distributed data processing.
- Extract user preferences and behavioral patterns.
- Apply collaborative filtering or machine learning algorithms.
- Generate personalized recommendations.
- Continuously update recommendations using new user behavior.

Example: If a user frequently watches science-fiction movies, the system recommends similar movies.

Distributed processing allows millions of user activities to be processed efficiently, while behavioral analytics improves personalization.

This increases **recommendation accuracy, user engagement and customer retention.**

8. Role of Cloud Computing in Big Data

Cloud computing provides scalable infrastructure and services for big data applications.

Advantages:

- **Scalability:** Resources can be increased or decreased according to workload.
- **Cost management:** Pay-as-you-use pricing reduces large upfront infrastructure costs.
- **Flexibility:** Organizations can select different storage and computing services.
- **Availability:** Cloud platforms provide highly available infrastructure.
- **Faster deployment:** Big data environments can be created quickly.
- **Managed services:** Cloud providers reduce infrastructure management requirements.

Limitations:

- Data security and privacy concerns.
- Dependence on internet connectivity.
- Vendor lock-in.
- Unexpected cloud costs.
- Compliance challenges.
- Data migration can be difficult.

Therefore, cloud computing provides **flexibility and scalability**, but organizations must carefully manage **security, cost and compliance.**

9. Real-Time Transportation and Logistics Analytics

A transportation system can use big data to improve route planning and operational decisions.

Framework:

GPS/IoT/Sensor Data → Data Ingestion → Real-Time Processing → Analytics → Route Optimization → Decision Making

- GPS devices continuously provide vehicle locations.
- IoT sensors collect fuel consumption, speed and vehicle conditions.
- Traffic and weather information are integrated.
- Kafka can collect real-time streaming data.
- Spark/Flink can process the data.
- Analytics models identify traffic congestion and delays.
- Route optimization algorithms select efficient routes.
- Dashboards provide real-time information to managers.

Benefits:

- Reduced travel time.
- Reduced fuel consumption.
- Better route planning.
- Reduced delivery delays.
- Improved fleet utilization.
- Faster operational decision-making.

Example: If heavy traffic is detected on the current route, the system can automatically recommend an alternative route.

10. Ethical and Legal Challenges of Big Data

Big data creates several ethical and legal concerns because organizations collect and analyze large amounts of personal information.

Ethical challenges:

- Privacy violations.
- Unauthorized data collection.
- Lack of user consent.
- Algorithmic bias.
- Discrimination.
- Excessive surveillance.
- Lack of transparency.

Legal challenges:

- Data protection requirements.
- Consent and purpose limitation.
- Data retention requirements.
- Cross-border data transfer.
- Unauthorized access and data breaches.
- Compliance with applicable privacy regulations.

Organizational policies:

- Obtain informed user consent.
- Collect only necessary data.
- Encrypt sensitive information.

- Apply role-based access control.
- Anonymize or pseudonymize personal data.
- Conduct regular security audits.
- Maintain clear data retention policies.
- Monitor AI models for bias.
- Provide transparency about data usage.
- Establish incident response procedures.